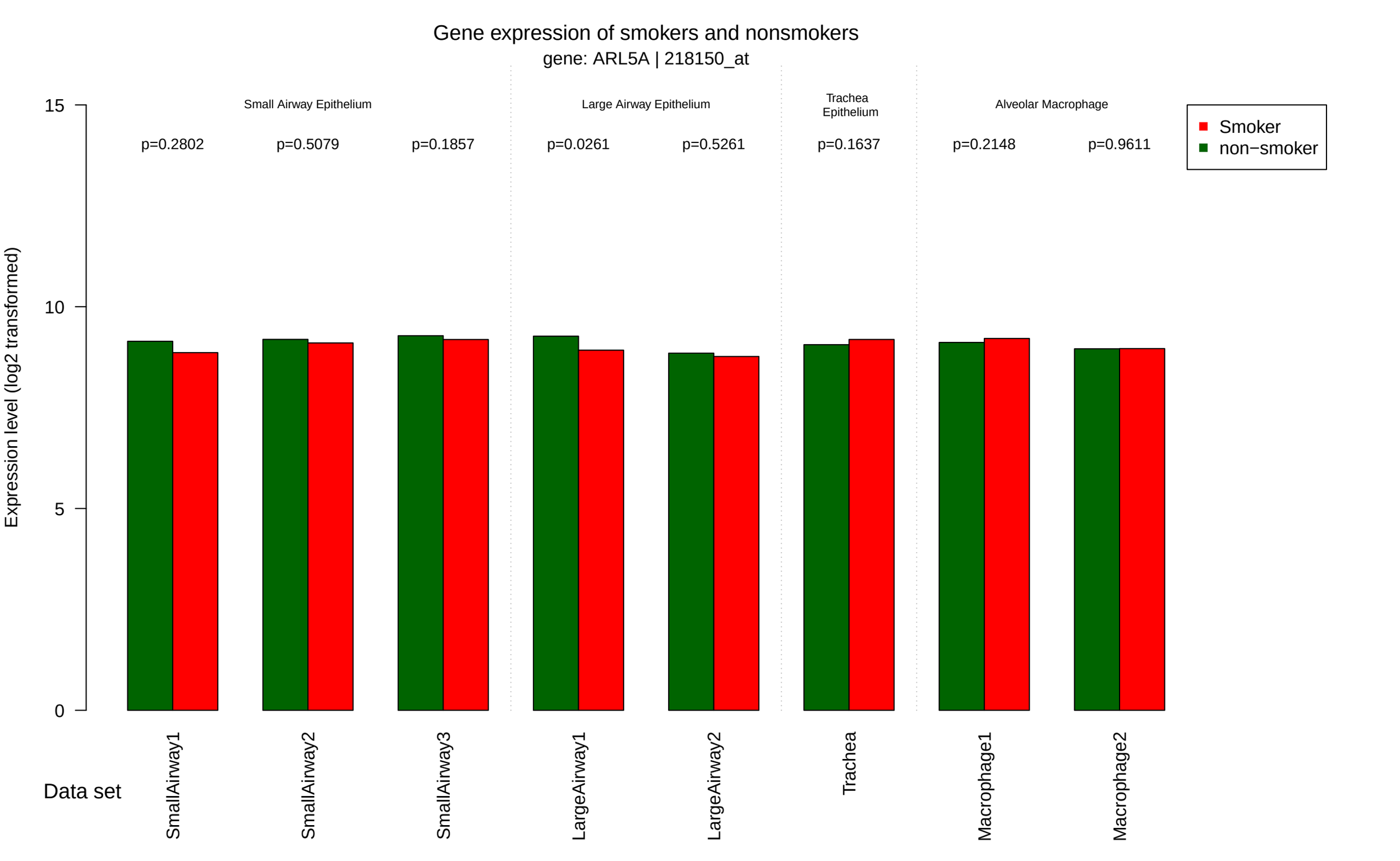
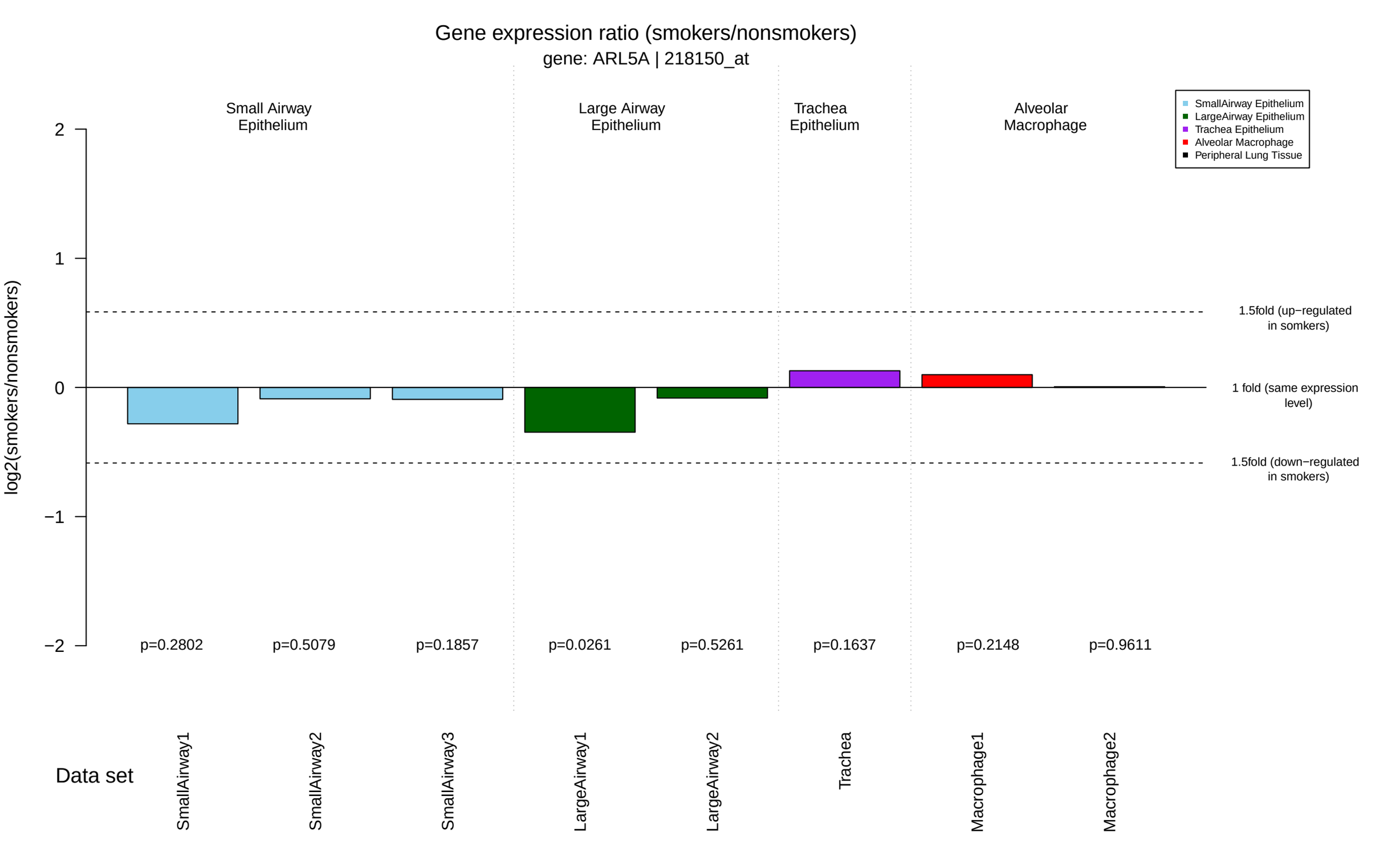
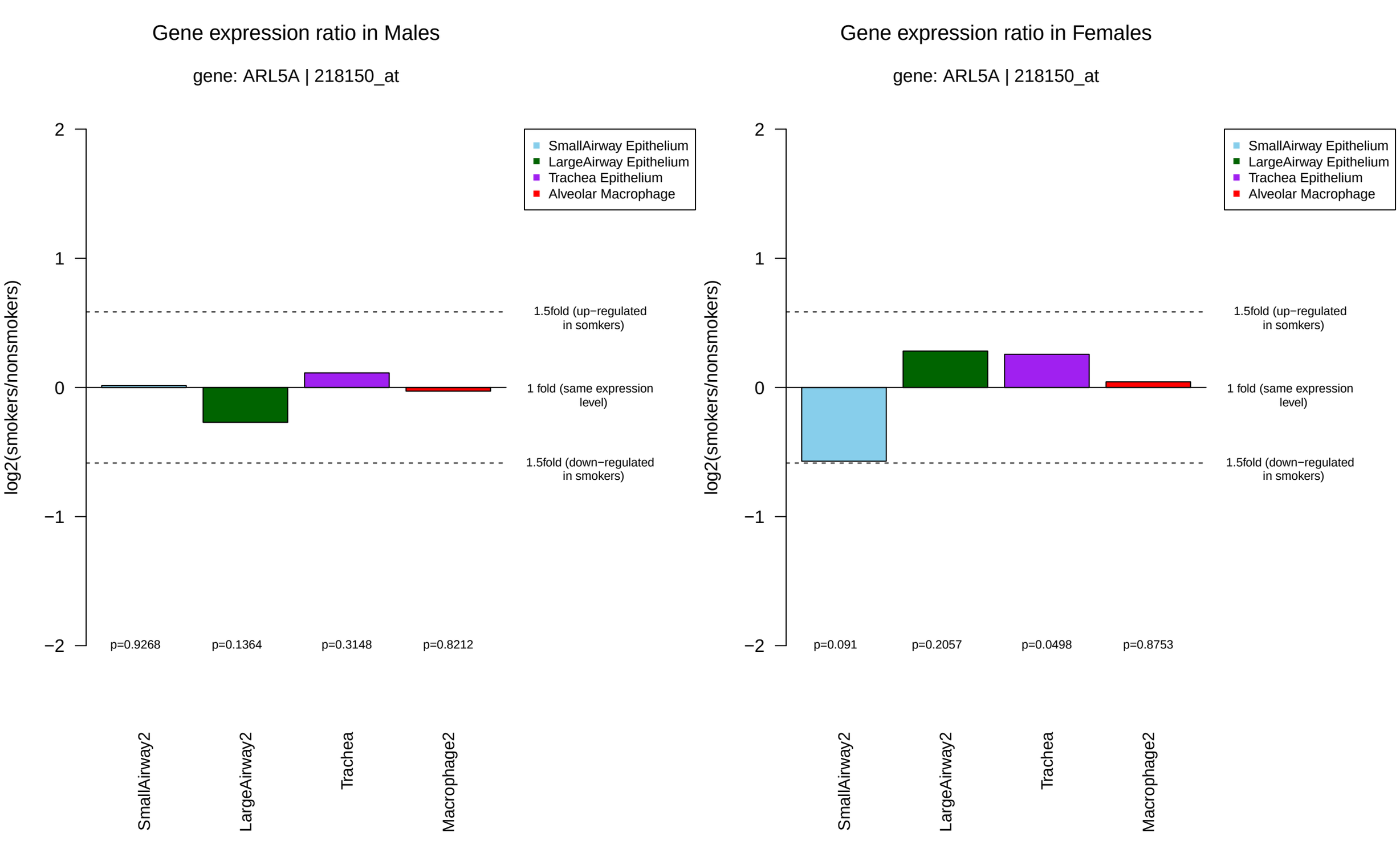
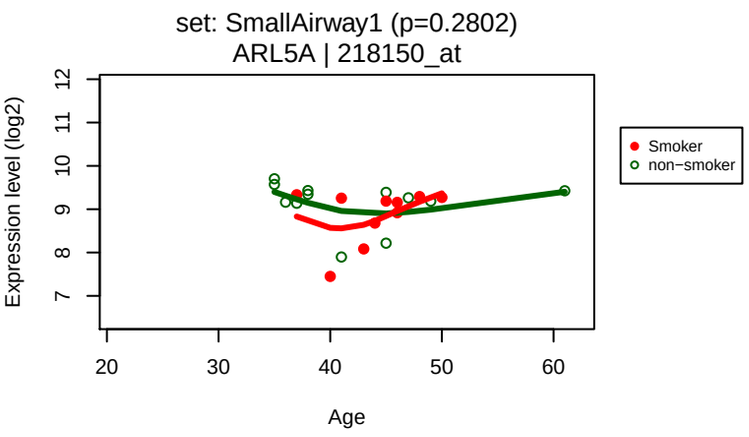
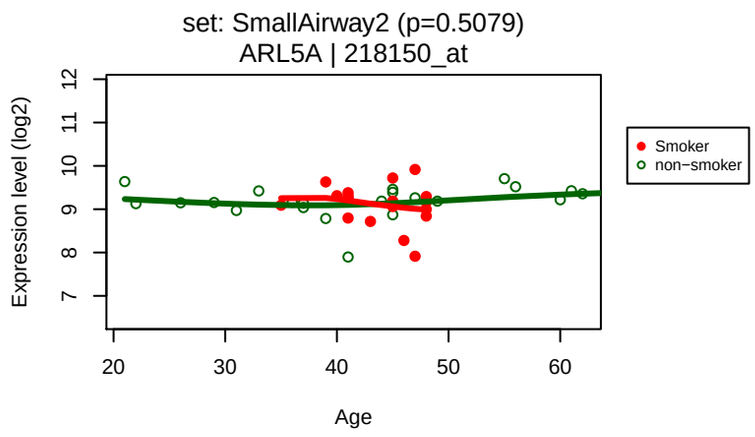
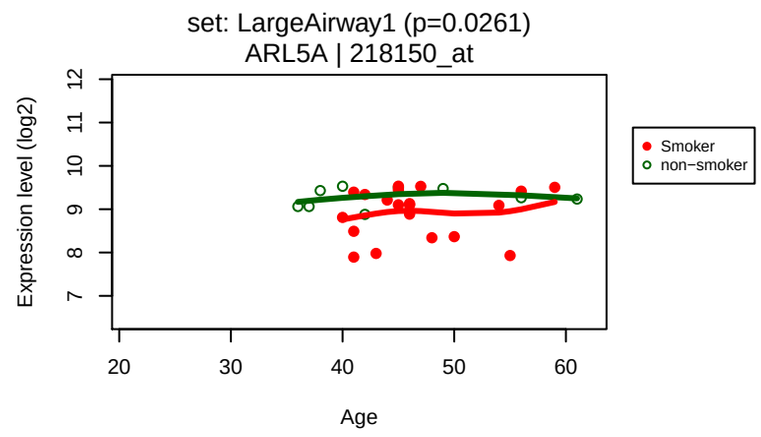

Gene expression of smokers and nonsmokers
gene: ARL5A | 218150_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2802
p=0.5079
p=0.1857
p=0.0261
p=0.5261
p=0.1637
p=0.2148
p=0.9611
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

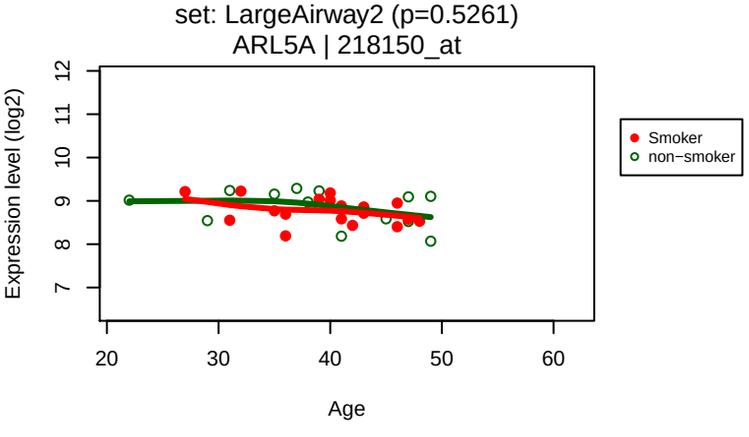

Gene expression ratio (smokers/nonsmokers)
gene: ARL5A | 218150_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2802
p=0.5079
p=0.1857
p=0.0261
p=0.5261
p=0.1637
p=0.2148
p=0.9611
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

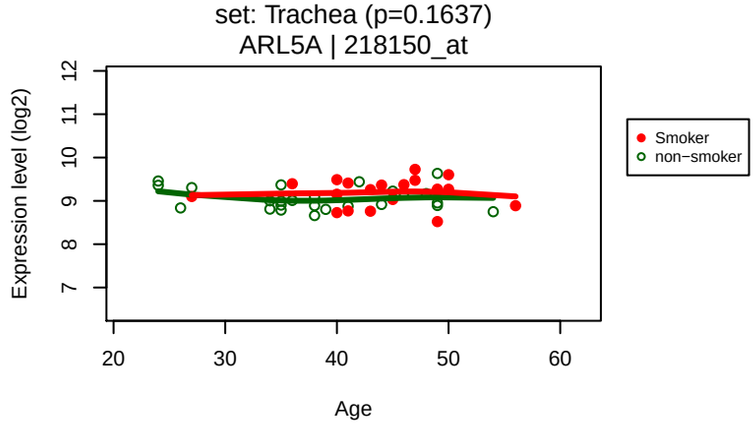

Gene expression ratio in Males
gene: ARL5A | 218150_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9268
p=0.1364
p=0.3148
p=0.8212
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL5A | 218150_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.091
p=0.2057
p=0.0498
p=0.8753
SmallAirway2
LargeAirway2
Trachea
Macrophage2

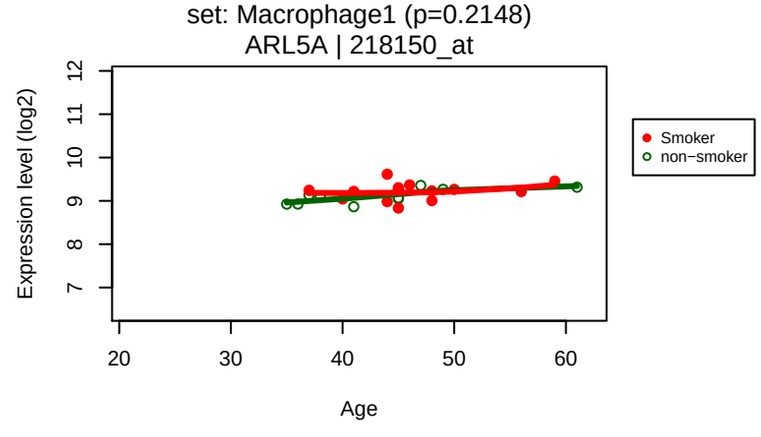

set: SmallAirway1 (p=0.2802)
ARL5A | 218150_at
Expression level (log2)
Age
Smoker
non−smoker

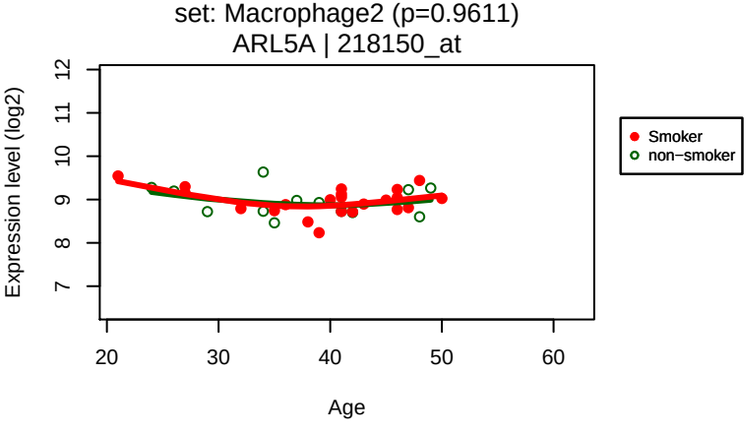

set: SmallAirway2 (p=0.5079)
ARL5A | 218150_at
Expression level (log2)
Age
Smoker
non−smoker

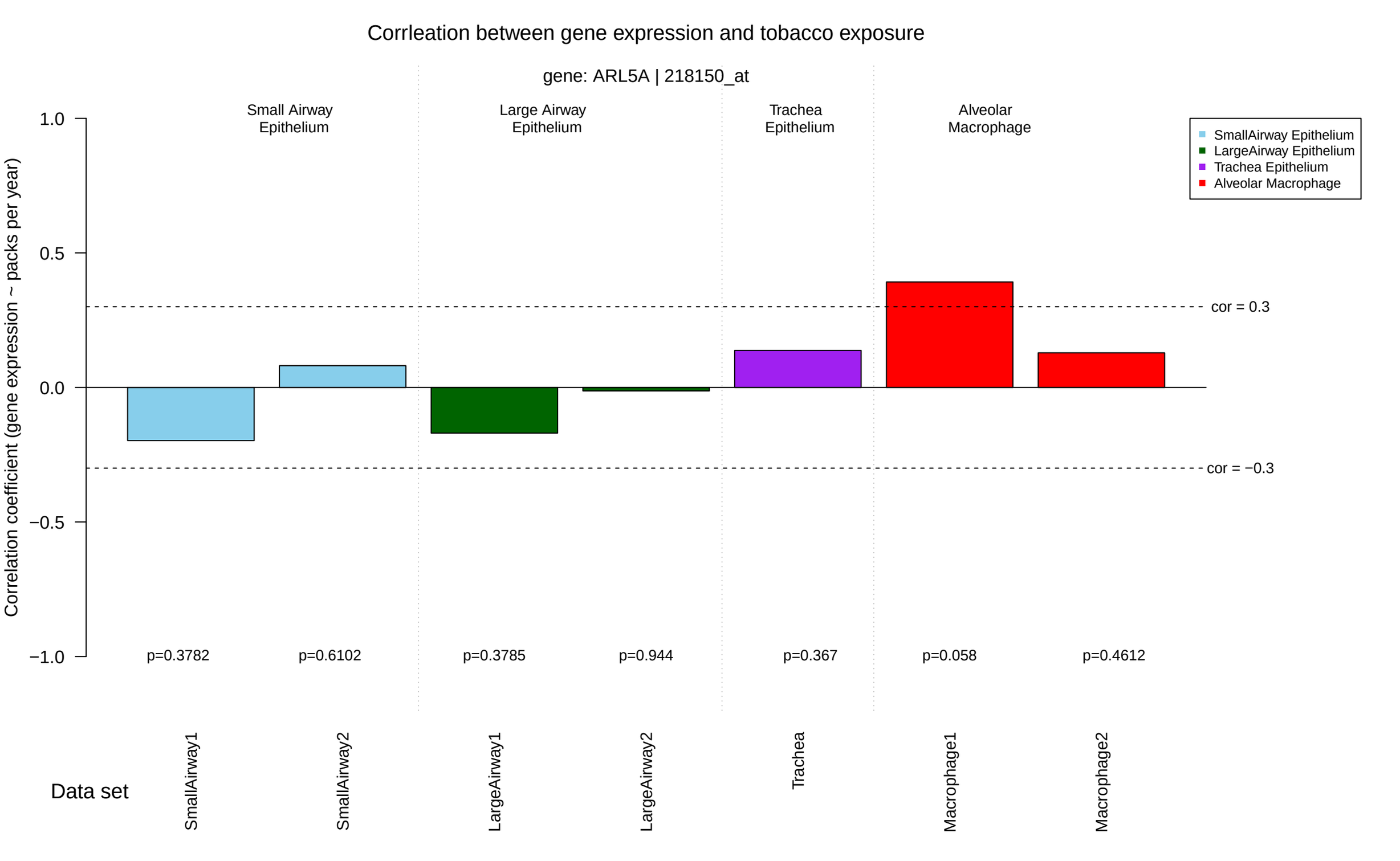

set: LargeAirway1 (p=0.0261)
ARL5A | 218150_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5261)
ARL5A | 218150_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1637)
ARL5A | 218150_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2148)
ARL5A | 218150_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9611)
ARL5A | 218150_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL5A | 218150_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3782
p=0.6102
p=0.3785
p=0.944
p=0.367
p=0.058
p=0.4612
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2